Eranda Sathsara Abewardhana

AI Engineer | Machine Learning Engineer | Full-Stack Developer

BSc (Hons) Artificial Intelligence, University of Moratuwa

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PROFESSIONAL SUMMARY

Completed Bsc hon in artificial intelligence at the University of Moratuwa with 2+ years of industry experience as a Machine Learning Engineering Intern and Mobile Application Developer. Proficient in designing multi-agent AI systems, Retrieval-Augmented Generation (RAG) pipelines, and production-grade REST APIs using FastAPI and Django. Experienced in cloud-native deployment on AWS (ECS, EC2, S3, Lambda) with Docker and CI/CD via GitHub Actions. Strong expertise in LLMs, NLP, PyTorch, LangChain, LlamaIndex, and the OpenAI Agent SDK, with full-stack capability across React, Angular, Next.js, and Flutter.

EDUCATION

University of Moratuwa **2022 – 2026**
BSc (Hons) in Artificial Intelligence

University of Colombo **2019 – 2020**
Certificate in Java Programming

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript (ES6+), TypeScript, C, C++, R

Backend: FastAPI, Django, Django REST Framework, Node.js, Express.js, .NET Web API

Frontend: React.js, Angular, Next.js, HTML5, CSS3, Tailwind CSS

Mobile: Flutter, Native Android (Java/Kotlin), React Native

Cloud and DevOps: AWS (ECS, EC2, S3, Lambda, IAM), GCP, Docker, Kubernetes, GitHub Actions, CI/CD

Databases: MySQL, PostgreSQL, MongoDB, MSSQL, Firebase, Pinecone (Vector DB)

AI and Machine Learning: TensorFlow, PyTorch, Scikit-learn, LangChain, LlamaIndex, OpenAI Agent SDK, Hugging Face, BERT

NLP and LLMs: RAG, Multi-Agent Systems, Prompt Engineering, Fine-tuning, Vector Embeddings, Semantic Search

Tools: Git, GitHub, Jira, Azure DevOps, Postman, PyCharm, IntelliJ

WORK EXPERIENCE

Machine Learning Engineering Intern **Feb 2025 – Aug 2025**
Orysys Pvt Ltd, Colombo, Sri Lanka

- Architected scalable backend microservices and REST APIs in FastAPI to serve AI model inference, document indexing, and workflow orchestration at production scale.
- Built end-to-end production-grade RAG systems and multi-agent AI workflows enabling intelligent document search, question answering, and automated reasoning over enterprise knowledge bases.
- Managed a Pinecone vector database for semantic search; trained time-series forecasting models using PyTorch, Prophet, and ARIMA, improving prediction accuracy on business KPIs.
- Collaborated in an Agile team using Jira and Azure DevOps; contributed to sprint planning, code reviews, and technical documentation.

Mobile Application Developer **Jan 2021 – Mar 2022**
eVision Microsystems Pvt Ltd, Colombo, Sri Lanka

- Developed and shipped 3+ cross-platform mobile applications using Flutter and Native Android (Java) for commercial clients, consistently meeting delivery deadlines.
- Integrated RESTful APIs via .NET Core Web API for real-time data synchronization, push notifications, and secure mobile-to-backend communication.
- Maintained 95%+ code quality through unit testing and peer code review, reducing post-release bug rates significantly.

PROJECTS

Yukti – Sri Lankan Legal AI Assistant **2025**
Multi-Agent AI, RAG, LlamaIndex, FastAPI, AWS ECS, React
Frontend: github.com/Eranda-Abewardhana/Yukthi-Fe Backend: github.com/Eranda-Abewardhana/Yukthi-Be

- Designed full system architecture for an AI-powered legal chatbot using LLMs to classify queries, retrieve statutory documents, and generate cited answers with hallucination mitigation via Guardrails AI.
- Built async FastAPI backend with Celery task queue; implemented LlamaIndex AutoMerging Retrieval for hierarchical legal document RAG; applied Tree of Thoughts prompting over Groq-hosted Llama 3B.
- Deployed on AWS ECS with GitHub Actions CI/CD; sessions stored in MySQL; frontend built with React (Vite).

Call Center Communication Analyzer (SentiView) **2024**
NLP, Sentiment Analysis, AWS, Angular, FastAPI, MongoDB
Frontend: github.com/HRS0986/iCMS-Frontend Backend: github.com/Eranda-Abewardhana/ICSMS-Call-Analyzer-Backend

- Built NLP-based sentiment analysis platform for call recordings; engineered audio ingestion and transcription pipeline using Angular and FastAPI.
- Leveraged AWS S3, Lambda, and EC2 for serverless audio processing and API hosting; Dockerized full stack with GitHub Actions CI/CD; metadata stored in MongoDB.

Deep Learning Framework for Genomic Variant Pathogenicity Prediction **2026**
Research, Deep Learning, Bioinformatics, Explainable AI, PyTorch, CNNs, Transformer Encoders

- Designed a novel dual-input architecture combining Multi-Scale CNNs and Transformer Encoders for REF/ALT genomic sequences (1024bp) to predict pathogenic impact of single-nucleotide variants (SNVs).
- Implemented in PyTorch with XAI via Integrated Gradients and In-silico Mutagenesis (Captum); evaluated using ROC-AUC, AUPRC, MCC, and Balanced Accuracy on gene-balanced ClinVar datasets.

YouFIRA – Autonomous Robotic Assistance Vehicle **2022–2023**
Computer Vision, Robotics, OpenCV, TensorFlow Lite, ROS, C++
Repository: github.com/Eranda-Abewardhana/Intelligence-Assistant

- Built an autonomous assistive vehicle using OpenCV for object detection and TensorFlow Lite for on-device inference; designed a robotic arm with PID Control and Inverse Kinematics.
- Integrated full embedded stack in C++ with ROS for sensor/actuator coordination; fabricated a custom PCB and companion voice-command mobile app.

ACHIEVEMENTS AND LEADERSHIP

- **8th Place** – Code Rush algorithmic competition, University of Moratuwa *2022*
- **Best Performing Local Board Member** – iGT Department, AIESEC UoM *2024/25*
- **Competitor** – IDIALIZE 2024, CodeSprint 7.0, MadHack 2023, ACES Coders v10.0 *2023–2024*
- **Active Member** – Rotaract Club and Classical Music Society, University of Moratuwa *Ongoing*

OTHER SKILLS

Languages: English (Professional), Sinhala (Native) **Soft Skills:** Leadership, Communication, Agile Teamwork, Problem Solving

REFERENCES

Prof. Subha Fernando, Professor in Artificial Intelligence, University of Moratuwa – subhaf@uom.lk

Mr. Janith Liyanage, Director, eVision Microsystems Pvt Ltd – janith@evisionmicro.com +94 718 543 880